NATIONAL UNIVERSITY OF MODERN LANGUAGES

Lab no 3

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PROGRAM: BSSE (6th Semester)

COURSE: AI Lab

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List Data Structure :

```python
# create list
numbers = [10, 20, 30, 40]

print("Original List:", numbers)

# add element
numbers.append(50)
print("After Append:", numbers)

# remove element
numbers.remove(20)
print("After Remove:", numbers)

# access element
print("First Element:", numbers[0])

# loop through list
print("List Elements:")
for n in numbers:
    print(n)
```

Output :

```
Original List: [10, 20, 30, 40]
After Append: [10, 20, 30, 40, 50]
After Remove: [10, 30, 40, 50]
First Element: 10
List Elements:
10
30
40
50
```

Stack (Last In First Out) :

```python
stack = []

# push operation
stack.append(10)
stack.append(20)
stack.append(30)

print("Stack after push:", stack)

# pop operation
stack.pop()

print("Stack after pop:", stack)

# top element
print("Top element:", stack[-1])
```

Output :

```
Stack after push: [10, 20, 30]
Stack after pop: [10, 20]
Top element: 20
```

Queue (First In First Out) :

```python
queue = []

# enqueue
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue after enqueue:", queue)

# dequeue
queue.pop(0)

print("Queue after dequeue:", queue)

# first element
print("Front element:", queue[0])
```

Output :

```
Queue after enqueue: [10, 20, 30]
Queue after dequeue: [20, 30]
Front element: 20
```

Linked List

```python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None


class LinkedList:
    def __init__(self):
        self.head = None

    def insert(self, data):
        new_node = Node(data)

        if self.head is None:
            self.head = new_node
            return

        temp = self.head
        while temp.next:
            temp = temp.next

        temp.next = new_node
```

```python
    def display(self):
        temp = self.head
        while temp:
            print(temp.data, end=" -> ")
            temp = temp.next


# create linked list
ll = LinkedList()

ll.insert(10)
ll.insert(20)
ll.insert(30)

print("Linked List:")
ll.display()
```

```
Linked List:
10 -> 20 -> 30 ->
```

Binary Tree :

```python
class Node:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None


# create nodes
root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

# inorder traversal
def inorder(node):
    if node:
        inorder(node.left)
        print(node.data, end=" ")
        inorder(node.right)

print("Inorder Traversal:")
inorder(root)
```

Output :

```
Inorder Traversal:
4 2 5 1 3
```

Graph:

```python
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

# display graph
for node in graph:
    print(node, "->", graph[node])
```

```
A -> ['B', 'C']
B -> ['D', 'E']
C -> ['F']
D -> []
E -> ['F']
F -> []
```